

AI-Powered Dynamic Task Allocation for Agile Work Environments

Overview:

This paper represents the use of artificial intelligence for dynamically allocating tasks in modern agile and remote-working environments. It basically focuses on using information about the employees, workload, skills and task requirements to improve the overall work or task process.

Project Objective:

The actual objective is to develop an AI-based task-allocation approach that can be similar or can match tasks with appropriate workers while considering changing workloads. The approach is intended to reduce inefficient task distribution and update or improve productivity and work completion.

Research Gap:

Traditional task-allocation systems and methods generally depend on fixed specific rules or manual decisions and may not respond effectively when workload, employee availability or task requirements change. There is therefore a need for an adaptive AI-based allocation system and methods.

Research Questions:

How can AI dynamically allocate tasks based or according to workers' skills and workload?
How can intelligent allocation update and improve task completion time, workload balance and productivity in agile working environments?

Key Contribution:

The study proposes an AI framework using techniques including natural-language processing and reinforcement learning. Its findings demonstrate how AI can support dynamic task allocation and provide a foundation for intelligent productivity-support systems.

IEEE

Interactions with Generative AI: Wearables to Measure Developer Experience and Productivity Objectively

Overview:

This research presents how wearable technology can be used to understand human experiences while interacting with Generative AI. It particularly considers productivity, cognitive load and physiological replies and responses during AI-supported work.

Project Objective:

The objective is to find whether physiological information collected through wearable devices can provide an objective measurement of user experience and productivity. This can complement traditional questionnaires and self-reported productivity measurements.

Research Gap:

Most productivity studies depend heavily on subjective surveys or simple measurements of completed work. These methods may fail to capture cognitive workload, stress and other changes occurring while users interact with AI systems.

Research Questions:

Can wearable physiological data measure cognitive load during Generative-AI interaction? Can combining physiological and subjective information provide a better understanding of productivity and user well-being?

Key Contribution:

The paper proposes combining wearable physiological measurements with subjective evaluations to study developer experience. This is particularly relevant to FocusBot because sensors could potentially monitor user states and allow an AI desk companion to provide appropriate productivity or break recommendations.

IEEE

The Smart Shift: Generative AI and the Rise of Human-AI Productivity

Overview:

This paper studies the increasing role of Generative AI in improvement on human productivity. It examines how AI can assist people with routine activities, data, information processing, automation and decision-support tasks.

Project Objective:

Basically the objective is to understand how Generative AI can complement human capabilities and develop efficiency rather than simply replace human activities. The research considers AI as a productivity-support technology.

Research Gap:

Although Generative AI is increasingly used in workplaces, there is still a need to understand how humans and AI can perform work together effectively. Existing discussions often emphasize automation without sufficiently investigating and examining human-centered productivity support.

Research Questions:

How can Generative AI improve human productivity? Which types of work can benefit from AI assistance while maintaining meaningful human involvement and decision-making?

Key Contribution:

The paper contributes a broader understanding of human-AI productivity and demonstrates the potential of AI for automation, analytics and decision support. These ideas provide theoretical

support for designing FocusBot as an assistant rather than an autonomous replacement for the user.

SpringerLink

Enhancing Knowledge Workers' Well-Being and Productivity: A Mindful Co-Working Design Approach

Overview:

This study focuses on knowledge workers who require long periods of concentration but also need communication and collaboration. The researchers introduce **Mindful Co-Working Design** . combining mindfulness, interruption management, breaks and goal-setting. The study involved **91 participants**.

Project Objective:

The objective is to determine whether a mindful working approach can improve productivity while also supporting workers' psychological well-being. It particularly addresses the toughness

of maintaining deep concentration in environments including continuous digital and social interruptions.

Research Gap:

Existing productivity approaches often focus on completing more work but may overlook stress, concentration and emotional well-being. There is a need for approaches that determine productivity and well-being together rather than treating them as different problems.

Research Questions:

Can mindful working practices improve concentration and productivity? Can they reduce stress and negative emotional experiences while people work collaboratively?

Key Contribution:

The study found improvements in concentration, goal accomplishment and well-being, with the mindful condition showing stronger reductions in results which are negative. This gives a useful foundation for FocusBot's **break alarms, focus sessions, interruption control and well-being features**.

SpringerLink

Intelligent Technology and Enhanced Well-Being: Can Artificial Intelligence Mitigate Digital Overload?

Overview:

This research studies the relationship between artificial intelligence, employee efficiency, digital overload and well-being. It is particularly relevant because excessive digital technology use can increase workload and create negative experiences even when technology improves efficiency.

Project Objective:

The objective is to understand whether intelligent technologies can develop work efficiency while also helping users manage the negative effects associated with digital overload. It considers the balance between technological benefits and employee well-being.

Research Gap:

AI research frequently emphasizes productivity and efficiency, while the possible consequences or results of excessive technology use receive comparatively decreased attention. More research is required to understand how AI can support productivity without creating additional digital pressure.

Research Questions:

Can AI improve employee efficiency while supporting well-being? How does digital or techno-overload influence the relationship between AI use and employee well-being?

Key Contribution:

The study provides evidence that AI adoption should be considered together with its effects on users rather than evaluated only through productivity. This supports FocusBot's idea of monitoring work intensity and encouraging appropriate breaks instead of continuously pushing users to work.

AI and Work Design: A Positive Psychology Approach to Employee Well-Being

Overview:

This paper focused on artificial intelligence from the view of work design and positive psychology. Instead of considering AI only as a productivity technology, it considers how AI-supported work can influence employees' positive psychological experiences.

Project Objective:

The objective is to connect AI implementation with work-design principles that support employee well-being. The research focuses on encouraging human-centered approaches in which technological improvements also consider users' psychological needs.

Research Gap:

Many AI workplace studies emphasize automation, efficiency and performance while giving comparatively less attention to positive psychological results. There is therefore a need for research connecting AI-supported work with employee well-being.

Research Questions:

How can AI-supported work be designed to promote employee well-being? What work-design factors should be considered when introducing AI into everyday working environments?

Key Contribution:

The study connects AI, work structure design and positive psychology to provide a more human-centered perspective on AI adoption. For FocusBot, this supports designing features around **motivation, positive feedback, achievable goals and healthy working habits**, rather than productivity alone.

Mitigating the Effect of AI Anxiety on Employees' Creativity: A Social Cognitive Perspective

Overview:

This paper basically focuses on employees' anxiety related to artificial intelligence and its possible influence on creativity. It considers how employees psychologically respond when AI becomes part of their working environment.

Project Objective:

The main purpose or objective is to understand factors that can reduce the negative effects of AI-related anxiety on employees. It emphasizes the importance of considering human attitudes and psychological responses when implementing the AI systems.

Research Gap:

AI workplace research often focuses on technical capability and productivity, while employees' emotional responses to AI may receive less attention. AI anxiety can potentially interfere with creativity and positive interaction with AI systems.

Research Questions:

How does AI-related anxiety affect employees' creativity? What social, organizational or personal factors can help reduce negative reactions toward AI technology?

Key Contribution:

The research provides a social-cognitive perspective for understanding human responses to AI. This is relevant to FocusBot because an interactive assistant should have a supportive and non-threatening interface rather than constantly reminding users about performance.

Intelligent Support for Digital Wellbeing: A Design Framework Through a Systematic Literature Review

Overview:

This 2025 ScienceDirect/Elsevier study systematically reviews AI-powered applications designed to support digital and emotional well-being. It examines how AI can analyze user behavior and provide personalized interventions.

Project Objective:

The actual goal is to identify important characteristics, challenges and opportunities in AI-based digital-wellbeing systems. The researchers then improve a design framework that developers can use when creating such technologies.

Research Gap:

Existing digital-wellbeing applications may not adequately address personalization, privacy, ethical concerns and user dependence on automated interventions. There is a need for structured guidance for developing responsible AI well-being systems.

Research Questions:

What are the main characteristics of AI-powered digital-wellbeing systems?

What design factors should developers consider regarding personalization, privacy, user data and intervention strategies?

Key Contribution:

The researchers propose a framework containing **6 major dimensions and 23 sub-dimensions**. The paper is highly relevant to FocusBot because it provides guidance for designing personalized, adaptive and privacy-conscious AI well-being features.

AI and Employee Wellbeing in the Workplace: An Empirical Study

Overview:

This 2025 Journal of Business Research study investigates how AI adoption affects employee well-being. The researchers analyzed data from **207 companies** using survey data and structural equation modeling.

Project Objective:

The objective is to understand whether AI directly improves employee well-being or whether its effects occur through improvements in work-related factors. The study specifically examines task optimization and workplace safety.

Research Gap:

The relationship between AI adoption and employee well-being remains relatively under-researched compared with AI's effects on productivity and efficiency. The mechanisms through which AI may influence well-being therefore require further investigation.

Research Questions:

Does AI adoption directly influence employee well-being? Do task optimization, workplace safety and security act as mechanisms through which AI affects employee well-being?

Key Contribution:

The study found that AI adoption did **not directly** affect employee well-being but influenced it indirectly through task optimization and safety. This supports FocusBot's concept that AI should improve the user's actual working process rather than simply provide more technology.

Unpacking Human-AI Interaction: Exploring Unintended Consequences on Employee Well-Being

Overview:

This 2025 Journal of Business Research paper examines the less obvious consequences of human-AI interaction in workplace environments. It looks beyond productivity and considers how AI-supported work can influence employees' psychological, physical and social experiences.

Project Objective:

The objective is to identify unintended consequences that may occur when AI becomes integrated into everyday work. The study aims to provide a more clear about gg understanding of human well-being during AI-supported activities.

Research Gap:

Much AI research emphasizes benefits such as automation, efficiency and productivity, while unintended negative consequences receive less attention. More research is therefore needed on the human side of AI adoption.

Research Questions:

What unintended consequences can emerge from human-AI interaction? How can these consequences influence employees' psychological, physical and social well-being?

Key Contribution:

The study identifies many important well-being concerns associated with human-AI interaction. For FocusBot, this suggests that the system should not simply maximize working time, it should also recognize signs of excessive workload and encourage healthier interaction with technology.